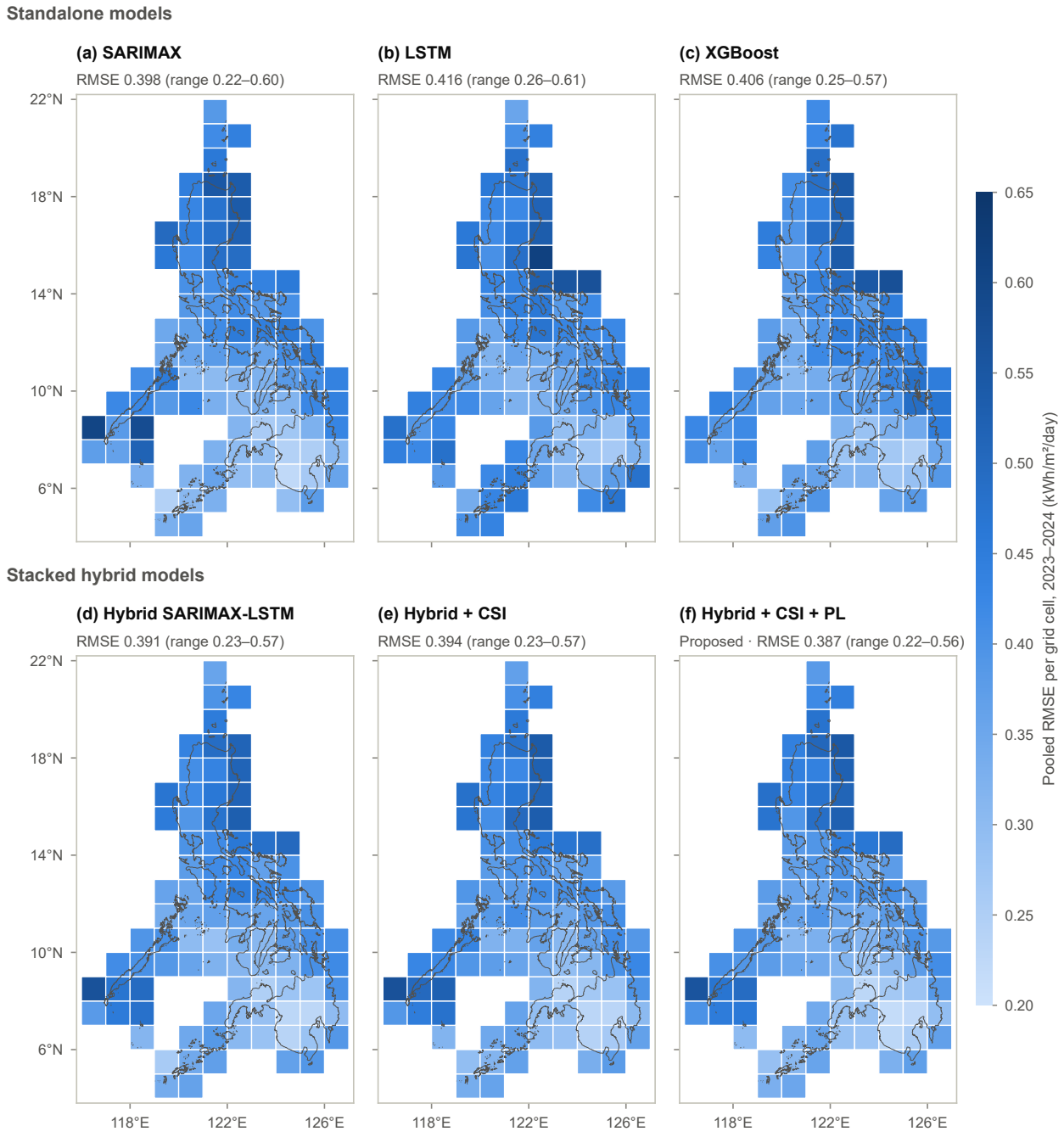

Standalone models
(a) SARIMAX
RMSE 0.398 (range 0.22–0.60)
(b) LSTM
RMSE 0.416 (range 0.26–0.61)
(c) XGBoost
RMSE 0.406 (range 0.25–0.57)
Stacked hybrid models
(d) Hybrid SARIMAX-LSTM
RMSE 0.391 (range 0.23–0.57)
(e) Hybrid + CSI
RMSE 0.394 (range 0.23–0.57)
(f) Hybrid + CSI + PL
Proposed · RMSE 0.387 (range 0.22–0.56)
22°N
18°N
14°N
10°N
6°N
118°E
122°E
126°E
Pooled RMSE per grid cell, 2023–2024 (kWh/m²/day)
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20